Mangesh Vishvnath Mundhe

ch25mtech11007@iith.ac.in | mangeshmundhe.vishvanath@gmail.com | +91 7028063728 | LinkedIn

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.46
Sharadchandra ji Pawar Junior college kodri Gangakhed Maharashtra	Class XII (Maharashtra State Board)	2018	68.77%
Mamata Vidyalaya Gangakhed Maharashtra	Class X (Maharashtra State Board)	2016	90.80%

POSITIONS OF RESPONSIBILITY

Chemistry Teacher (Online) | Prayatna Learning Centre, Bengaluru, Karnataka 2025-26

Provided one-to-one online Chemistry tutoring. Guided students in concept learning, problem-solving, and exam preparation. Duration: 6 Months.

Chemistry Teaching Intern | Xavier School, Karauli, Rajasthan 2022

Assisted in Chemistry teaching and student mentoring. Supported lesson planning, doubt solving, and exam preparation. Duration: 2 Months.

Indian Knowledge Systems (IKS) Students Chapter, IIT Hyderabad

Active Member -- Participated in initiatives promoting Indian heritage, traditional knowledge, and interdisciplinary learning.

Teaching Assistant (TA), Department of Chemical Engineering, IIT Hyderabad Aug 2025 - Present

Served as TA for three undergraduate courses(including Chemical reaction engineering, Thermodynamics and Heat transfer operations laboratory), mentoring 100+ students (till now), assisting the faculty in conducting tutorials, preparing question papers, evaluating assignments and exams, and clarifying core subject/Lab concepts to strengthen students' academic understanding.

SKILLS

Technical Software - Aspen Plus, Chemdraw, MS Excel

Analytical Techniques - XRD, BET, NH3-TPD, H2-TPR, SEM, TEM , FTIR, XPS, GC-MS, GC-FID analysis

RELEVANT COURSES

Advanced Mineral Processing

- Studied mineral beneficiation and separation techniques for processing complex ores, including flotation, gravity, and magnetic separation.
- Covered comminution, size classification, and particle-surface interactions in mineral processing.
- Studied ore characterization and process selection for efficient mineral separation.
- Gained insight into flowsheet design, optimization, and factors affecting recovery, grade, and separation efficiency.

Heterogeneous Reaction Engineering

- Studied catalytic kinetics, adsorption, diffusion, and heterogeneous reactor design

Physicochemical Fundamentals for Chemical Engineers

- Completed a fully student-led, presentation-based course through independent study.

- Prepared and delivered technical presentations on physicochemical topics, including light scattering.
- Explained complex concepts using structured explanations, equations, and examples.
- Developed skills in scientific communication, presentation, and peer teaching.

PROJECTS

One-Pot Conversion of Eucalyptus Oil to Sustainable Aviation Fuel Range Hydrocarbons

- Developing a low-cost bifunctional Ni/CeO-AlO catalyst for one-pot hydrodeoxygenation and hydrogenation of 1,8-cineole to SAF-range hydrocarbons.
- Synthesizing the catalyst using Evaporation-Induced Self-Assembly (EISA) and studying its physicochemical properties using BET, NH-TPD and H-TPR.
- Evaluating catalytic performance in a batch reactor and analysing products using GC-MS and GC-FID.

Aspen Plus Simulation and Process Design of Benzyl Acetate Production

- Developed a complete Aspen Plus process simulation for benzyl acetate production via esterification.
- Modelled RCSTR reactor, separators, recycle streams, flash unit, and vacuum distillation column.
- Optimized reaction and separation conditions to achieve high product purity.
- Evaluated process utilities and overall process performance.

Extraction of Natural Dyes from Plant Sources 2022

- Extracted natural dyes from plant-based materials using suitable solvent extraction techniques.
- Investigated the use of alum as a mordant to improve dye fixation and color intensity on textile fabrics.
- Optimized key process parameters such as extraction temperature, extraction time, and solvent-to-solid ratio.
- Evaluated the extracted dyes for their potential application as eco-friendly alternatives to synthetic dyes.

PERSONAL PURSUITS

Cooking**,** Cat Care & Interaction and Teaching & Mentoring